Read, Not Reasoned? Ground-Truth Causal Interventions on Multi-Hop Reasoning Chains

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Abstract

When a language model writes a reasoning chain, does its answer depend on that chain, and through which path? We study this with interventions whose ground truth is known by construction: inserted redundant hops (true necessity 0), programmatic input changes (true effect 0), and an exact mediation analysis in which the chain is set directly. We report four results on two multi-hop datasets with five models from two families (Qwen2.5 0.5B–7B, SmolLM2-1.7B). **(i) Step masking is blind to bridge reasoning.** Masking one intermediate hop changes p(gold) by at most 6.5 points, whereas masking it together with its descendants in the chain's reasoning graph changes it by 41–77 points; we show this is an identity, not noise (single-step masking estimates a controlled direct effect, which vanishes on bridge chains). **(ii) Resampling-based necessity with a fixed gold baseline is biased by the generator.** Probability-of-necessity estimates computed against the original chain assign 56–78 points of necessity to provably redundant steps; the bias equals one minus the rate at which the model can regenerate a correct continuation, and a resampled baseline removes it (-0.3 to 4.3 points). **(iii) Chains are read, not reasoned over.** Answers depend on the chain when the answer is a string written in it (41–77 points) and barely when it must be composed from it (-1.5 to 7.3 points at every scale). A binary intervention that appends a *wrong* stated answer is followed 46–67% of the time up to 3B—as often on questions the model solves from the plain chain as on those it does not—while Qwen2.5-7B follows it on only 10% of the questions it can solve. **(iv) Distractors harm chain-of-thought through the chain.** With the chain held at its original value, the effect of four distractor paragraphs on CoT vanishes (direct effect -0.6 to $+1.0$ EM, n=500), while direct answering is harmed directly (-3.8 to -4.6). A readout-level decoding probe that contrasts the model with interventions on its own hops therefore cannot help, and does not. **(v) Executing the composition outside the model does.** When the model only extracts facts and genders and a rule table induced from training data composes them along the kinship graph, CLUTRR accuracy rises by 21–37 points over compute-matched self-consistency on short chains (1.5B–7B, two unseen splits) and by 24 points on unseen long chains at 7B; given the same extracted facts, the model itself is no better than CoT. All hypotheses were fixed before each GPU run.

1 Introduction

Chain-of-thought prompting [30] makes models write intermediate steps, but writing a step is not the same as *using* it. Causal tests show that models frequently ignore or post-rationalise their chains [7, 14, 26, 27], and that step importance depends on how a step is removed [2, 11]. What these studies lack is ground truth: without knowing which steps are necessary, an estimator of step importance cannot be validated, only compared with other estimators.

We build that ground truth. On CLUTRR [20] (kinship graphs) and MuSiQue [25] (real multi-hop questions with gold decompositions) we write gold reasoning chains, insert one true but ir-

relevant hop, and ask each estimator to separate necessary from redundant hops. We pair this with binary input interventions whose effect on the answer is zero by construction, placebo and mask-token controls, and an exact mediation analysis in which the chain itself is set, so that direct and indirect effects are observed per question rather than estimated.

Contributions.

1. **A ground-truth protocol for chain reliance**: necessity labels and null effects known by construction; placebo, mask-marker and volume-matched controls; exact NDE/TIE mediation.
2. **Two estimator biases, with closed forms.** Single-step masking measures a controlled direct effect that vanishes on bridge chains (proposition 1); resampling necessity against a fixed gold continuation adds $1 - r$, where r is the model's regeneration success (proposition 2). Both are confirmed at every scale we test.
3. **Read, not reasoned.** Reliance on the chain appears when the answer can be copied from it and is near zero when it must be composed; a single binary intervention (T_{copy}) switches it on. Its wrong-answer variant separates copying from the inability to compose: up to 3B a stated wrong answer overrides the model's own solution; at 7B it is mostly vetoed, yet the relational hops still carry no measurable effect.
4. **Where robustness failures live.** Distractors harm chain-of-thought entirely through the regenerated chain, which explains why readout-level contrastive decoding—including a graph-aware rule we formulate as a probe (HGCD)—cannot help at this scale.
5. **A decoding intervention the diagnosis predicts.** Moving the composition out of the model (extract facts, execute the reasoning graph's structural equations with a rule table) beats compute-matched self-consistency by 13–37 points, and an ablation holding the extracted facts fixed shows the gain is caused by the composition step (section 7).

2 Related Work

Faithfulness of reasoning chains. Lanham et al. [7] intervene on model-written chains (truncation, inserted mistakes, paraphrases); Turpin et al. [26] show biased features steer answers without being verbalised; FRODO [14] uses causal mediation [28] and trains small models with counterfactual preferences; FUR [27] measures faithfulness by unlearning steps; Thought Anchors [2] measures step importance by counterfactual resampling, and Thought Branches [11] argues that step effects must be measured by resampling the continuation (a total effect), noting that removing a sentence while preventing its content from reappearing changes conclusions. FRODO already treats the rationale as a mediator and finds weak indirect effects for models under 20B, and Somov et al. [21] find that models often fail to update after edits to intermediate structures. Our contribution relative to these is narrower and specific: (i) necessity labels and null effects that are *known by construction*, so step-importance estimators can be *validated* rather than compared with each other; (ii) a deterministic, rollout-free total-effect intervention for hop-structured chains, validated against resampling; (iii) the bridge-versus-compositional distinction, which predicts exactly where direct-effect estimators fail; and (iv) an exact mediation of distractor effects (chain vs. readout).

Necessity and sufficiency. Yu et al. [32] define step-level probabilities of necessity and sufficiency for chain-of-thought, prove identifiability under exogeneity and monotonicity (building on

Table 1: Positioning against closely related work. "GT" = the true value of the measured effect is known by construction.

Work	What is intervened on	Effect measured	GT?	Training	Controls
Lanham et al. [7]	model's CoT (truncate, add mistake, paraphrase)	answer change	no	none	paraphrase, filler tokens
Stolfo et al. [22] (ACL)	input factors of math problems	total / direct causal effect	partly	none	–
Paul et al. [14] (EMNLP-F)	rationale	mediation (indirect effect)	no	preference training	–
Yu et al. [32] (NeurIPS)	CoT step, replaced + rolled out	PNS	no (LLM judge)	SFT / ICL	–
Macar et al. [11] (ICLR)	sentence, resampled continuation	total effect, resilience	no	none	on- vs. off-policy
Somov et al. [21]	intermediate structure (rubric)	decision update rate	yes (deterministic map)	none	–
Tutek et al. [27] (EMNLP)	step information in weights	prediction change	no	unlearning	–
Shi et al. [19], Wang et al. [29]	whole context	(implicit) total context effect	no	none (decoding)	–
Ours	hop + DRG descendants; binary input interventions	direct vs. total effect; NDE/NIE	**yes**	none	placebo, mask-marker, redundant hop, content-free

Pearl [16], Tian and Pearl [24]), and use PNS to prune chains for in-context learning and fine-tuning. We show the same quantity arises as the limit of a decoding rule, and we separate direct from total effects [15].

Contrastive and guided decoding. Contrastive decoding [9, 13], DoLa [4], CFG [18], CAD and its adaptive/gated variants [19, 23, 29] contrast distributions under different models, layers or contexts. Visual contrastive decoding [8] uses a *distorted* rather than empty input; our L1 variant is the textual analogue with a type-matched counterfactual. Pced [5] treats each retrieved document as an expert contrasted with a no-context baseline; RED [31] multiplies image- and rationale-conditional distributions in vision-language models. None of these intervene on the hops of the model's own reasoning graph, distinguish direct from total effects, or relate the contrast to PNS.

Causal analyses of reasoning. Stolfo et al. [22] quantify robustness of mathematical reasoning through interventions on a causal graph of the input; Zečević et al. [33] argue models "talk" causality without being causal; text-intervention effect estimation [6] uses LLM rewrites as treatments. Our interventions are programmatic, so their ground-truth effects are known exactly.

3 Setup: a Structural Causal Model of Multi-Hop Reasoning

Variables. A question Q, a context $P = (P_{\text{sup}}, P_{\text{dist}})$ of supporting and distractor passages, a chain of hops $H_1, \ldots, H_k$, and an answer A. Each hop is rendered as `Hop` i`:` ⟨`body`⟩ `=>` ⟨`result`⟩. The model acts as a *reader* mechanism $\pi(a \mid Q, P, h_{1:k})$ over a finite candidate set $\mathcal{A}$.

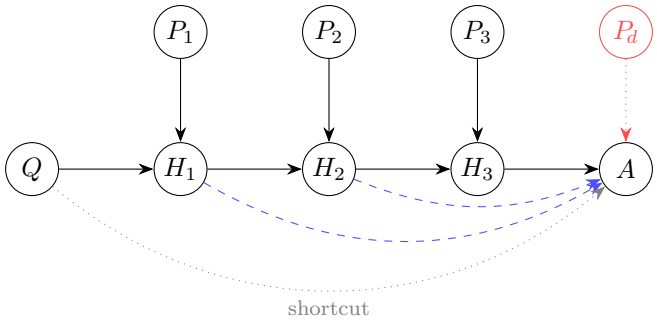


Figure 1: Task SCM. Solid: bridge chain. Dashed blue: extra edges of compositional chains. Red dotted: spurious distractor path that the *reader* may use but the task graph forbids. Gray: parametric shortcut. CAD's contrast includes the red path; HGCD holds P fixed and contrasts interventions on H_j only.

Two task graphs. In *bridge* chains (MuSiQue), hop j resolves an entity that the next hop consumes and the answer is the result of the last hop: $Q \to H_1 \to \cdots \to H_k \to A$, with $P_{\text{sup},j} \to H_j$. In *compositional* chains (CLUTRR), every hop contributes a relation that is composed into the answer: $H_j \to A$ for all j and $H_j \to H_{j+1}$ through the shared entity. Distractors have no edge to A in either task graph (fig. 1).

Directed reasoning graph (DRG). Given a generated chain $y = (h_1, \ldots, h_k)$ we add an edge $i \to j$ $(i < j)$ if the result of h_i is mentioned in the body of h_j. Terminal hops are those whose result is the query target (CLUTRR) or the last hop (MuSiQue); the *relevant set* $\mathcal{R}(y)$ contains the terminal hops and their DRG ancestors. Hops outside $\mathcal{R}(y)$ have no directed path to the answer.

Interventions on hops.

Definition 1 (Naive and graph-aware interventions)**.** y_j^{nai} *replaces hop* j *by a mask token and leaves all other hops unchanged.* y_j^{tot} *masks hop* j *and, in every DRG descendant* d *of* j*, masks every mention of the results of* j *and of* d*. The latter is a deterministic, rollout-free approximation of* $\text{do}(H_j{=}\perp)$*: descendant values are functions of* h_j*, hence unknown under the intervention. A* placebo *rewrites hop* j *without changing its entities or relation.*

For an answer a we write $\Delta_j^{\text{nai}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{nai}})$ and $\Delta_j^{\text{tot}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{tot}})$, suppressing Q, P.

What is (and is not) causal here. We do not assume the language model *is* an SCM of the task. The reader π is a deterministic function of its input, so editing hops in the prompt is an exact intervention on the reader's input, and Δ_j is an exact (not estimated) effect *on the reader*. The task SCM only supplies ground truth: which hops are necessary, and which input changes should have no effect on the answer.

Mask-token artifacts. Replacing text by `[MASKED]` is itself out of distribution. Our estimator benchmark controls for this: the inserted redundant hop is masked in exactly the same way, so any effect caused by the mask token alone appears as a non-zero Δ on redundant hops and lowers the AUC. In addition, a *mask-marker* control inserts `[MASKED]` into hop j while keeping all of its content; its effect Δ_j^{marker} measures the mask-token artifact directly and should be ≈ 0. Because

y_j^{tot} masks a whole subtree, we also use a *volume-matched* control that inserts the same mask tokens into hop j and every descendant while keeping all of their content, and a counterfactual-replacement version of the total effect that uses no mask on hop j at all.

4 Theory: two estimator biases

Assumption 1 (Markov reader on bridge chains)**.** *For bridge chains, $\pi(a \mid h_{1:k})$ depends on $h_{1:j}$ only through $h_{j+1:k}$ whenever h_{j+1} restates the result of h_j $(j < k)$.*

Proposition 1 (Single-step masking is a direct effect)**.** *$\Delta_j^{\text{nai}}(a)$ equals the controlled direct effect of H_j on A with $H_{j+1:k}$ held at their observed values. Under assumption 1, $\Delta_j^{\text{nai}}(a) = 0$ for every $j < k$ on bridge chains, whereas $\Delta_j^{\text{tot}}(a)$ is not constrained to zero.*

Proposition 2 (Fixed-baseline resampling is biased by the generator)**.** *Let q_j be the probability of a correct answer when hop j is corrupted and the continuation is resampled, and r_j the same probability when hop j is kept and the continuation is resampled. On chains answered correctly with the original continuation ($\text{PS} = 1$), the fixed-baseline estimator $1 - q_j$ equals $(r_j - q_j) + (1 - r_j)$: the resampled-baseline effect plus a bias $1 - r_j \geq 0$ that does not depend on whether hop j is necessary. For a redundant hop ($r_j = q_j$) it returns $1 - r_j$ instead of 0.*

Remark. The identifiability lemma of Yu et al. [32] assumes $P(A{=}y \mid \text{do}(S)) = 1$, i.e. $r_j = 1$; proposition 2 quantifies what happens when the rollout model cannot regenerate a correct continuation. Thought-anchor style resampling [2] uses a resampled baseline and is unbiased in this sense.

5 Experimental Protocol

Data. CLUTRR (`gen_train234_test2to10`), restricted to clean simple-path stories (11,708 train / 635 test / validation), gives exact gold paths. **MuSiQue-Ans** gives real text with gold decompositions; the context holds the supporting paragraphs plus 4 distractors. MuSiQue was built to resist single-hop shortcuts [12].

Estimator benchmark. For each gold chain we insert one true but off-path hop (necessity 0) and score every hop with: mask hop only (Δ^{nai}); mask hop plus DRG descendants (Δ^{tot}); counterfactual replacement of the hop's result without and with propagation; resampling necessity with a fixed gold baseline and with a resampled baseline ($K = 2$ samples per arm); and three controls (placebo rewrite, mask marker on the hop, mask markers on the whole subtree). Δ is measured on the gold answer's probability normalised over the candidate set and, as a robustness check that needs no candidate set, on its raw log-probability. Per model we use 180 CLUTRR and 150 MuSiQue gold chains (about 2,800 scored hops including both T_{copy} variants); for Qwen2.5-7B, 120 and 100 chains and no resampling estimators.

Copy vs. compute. $T_{\text{copy}} \in \{0, 1\}$ appends to a CLUTRR gold chain one hop that states the answer relation; everything else is identical. A second variant states a *wrong* relation that contradicts the relational hops; we report how often the model follows it, separately for chains the model solves without the extra hop.

Input interventions and mediation. `rename`, `order`, `distract` (CLUTRR) and `dist8`, `order` (MuSiQue) have true effect 0. For each question we observe $Y(0, M(0))$ (original input, original chain), $Y(1, M(1))$ (intervened input, regenerated chain) and $Y(1, M(0))$ (intervened input, *original* chain fed back), giving the natural direct effect $\text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]$ and the total indirect effect $\text{TIE} = \text{TE} - \text{NDE}$ [15] without cross-world assumptions.

Models and statistics. Qwen2.5-0.5B/1.5B/3B/7B-Instruct [17] and SmolLM2-1.7B-Instruct [1], fp16, greedy decoding, one Kaggle T4 each (7B split over two). Mediation is run on 500 MuSiQue questions for Qwen2.5-1.5B and 3B, and on 120 for the two full decoding runs. Bootstrap 95% CIs (2,000 resamples; for position means we resample questions, not hops), paired bootstrap and exact McNemar tests.

6 Results

All tables are produced by the released code; nothing is typed by hand.

Table 2: Effects on p(gold) by hop position across scale (points; estimator benchmark, means over hops, `tools/analyze_reliance_v2.py`). Question-clustered 95% CIs are in `positions_ci.csv`; e.g. CLUTRR non-final total at 3B is 7.3 [3.8, 11.0].

Quantity (Δp(gold), points)	Q-0.5B	Q-1.5B	Smol-1.7B	Q-3B	Q-7B
CLUTRR, non-final hop: total	0.1	-1.5	1.8	7.3	1.3
CLUTRR, non-final hop: placebo	-0.9	-0.4	-1.4	-1.4	-1.7
CLUTRR, final hop: total	3.0	-0.6	1.0	5.5	2.9
CLUTRR, final hop: placebo	0.7	-0.0	4.1	7.4	0.9
CLUTRR+copy, non-final: total	60.8	76.1	69.5	77.0	51.9
CLUTRR+copy, non-final: direct	1.3	1.3	1.0	1.7	0.9
CLUTRR+copy, non-final: marker-tree	11.0	4.0	6.8	3.6	3.7
MuSiQue, non-final: total	56.7	56.9	66.8	49.1	41.3
MuSiQue, non-final: direct	-2.4	0.6	-3.9	0.5	6.5
MuSiQue, non-final: marker-tree	7.3	-0.0	2.3	-0.3	5.7
MuSiQue, redundant: total	-4.5	0.4	-1.2	-1.5	-1.6
Rollout, gold baseline: necessary	79.0	75.7	68.9	61.5	–
Rollout, gold baseline: redundant	78.0	76.3	72.7	56.3	–
Rollout, resampled baseline: necessary	3.5	6.4	-0.2	11.3	–
Rollout, resampled baseline: redundant	1.7	-0.3	3.3	4.3	–

R1: single-step masking misses bridge reasoning (proposition 1). Wherever the answer is written in the chain (MuSiQue; CLUTRR with T_{copy}), masking one non-final hop moves p(gold) by -3.9 to $+6.5$ points at every scale, while masking it with its descendants moves it by 41–77 points (table 2). Total-effect estimators reach AUC 0.86–0.98 on MuSiQue and 0.94–1.00 on CLUTRR+copy, direct-effect ones 0.64–0.82 (table 3). The counterfactual-replacement implementation, which puts no mask on hop j, and the raw log-probability version, which needs no candidate set, agree with the masking one. The placebo control is at chance. The volume-matched control (mask tokens over the same subtree, content kept) moves p(gold) by -0.3 to 11 points against 41–77 for the total effect, so the effect is not an artefact of mask volume; it does carry some ranking

Table 3: Estimator benchmark across scale: AUC for separating necessary from redundant hops.

Dataset	Estimator	Q-0.5B	Q-1.5B	Smol-1.7B	Q-3B	Q-7B
MuSiQue	total	0.943	0.977	0.980	0.970	0.863
MuSiQue	direct (naive)	0.720	0.816	0.659	0.761	0.706
MuSiQue	cf, propagated	0.941	0.985	0.982	0.974	0.866
MuSiQue	total, raw log-prob	0.991	0.993	0.995	0.980	0.877
MuSiQue	rollout, gold base	0.511	0.524	0.505	0.551	–
MuSiQue	rollout, resampled	0.646	0.668	0.610	0.640	–
MuSiQue	marker-tree (control)	0.653	0.567	0.563	0.679	0.549
MuSiQue	placebo (control)	0.509	0.480	0.502	0.470	0.543
CLUTRR+copy	total	0.996	1.000	0.995	0.991	0.945
CLUTRR+copy	direct (naive)	0.637	0.666	0.647	0.703	0.724
CLUTRR+copy	cf, propagated	0.999	1.000	0.998	0.998	0.915
CLUTRR+copy	total, raw log-prob	0.995	1.000	0.996	0.988	0.962
CLUTRR+copy	marker-tree (control)	0.949	0.609	0.815	0.779	0.640
CLUTRR+copy	placebo (control)	0.560	0.469	0.385	0.467	0.462
CLUTRR	total	0.632	0.498	0.570	0.574	0.515
CLUTRR	direct (naive)	0.621	0.499	0.547	0.518	0.520
CLUTRR	cf, propagated	0.558	0.563	0.546	0.554	0.550
CLUTRR	total, raw log-prob	0.645	0.467	0.558	0.566	0.526
CLUTRR	rollout, gold base	0.501	0.496	0.503	0.504	–
CLUTRR	rollout, resampled	0.510	0.486	0.497	0.535	–
CLUTRR	marker-tree (control)	0.508	0.534	0.501	0.457	0.474
CLUTRR	placebo (control)	0.511	0.483	0.534	0.522	0.571

signal on CLUTRR+copy (AUC up to 0.95 at 0.5B), which we report as a limitation of mask-based controls for the smallest model.

R2: fixed-baseline resampling assigns necessity to redundant steps (proposition 2). On MuSiQue the fixed-baseline resampling estimator gives 56–78 points of necessity to the *redundant* hop, indistinguishable from necessary hops (61–79; AUC 0.51–0.55). By proposition 2 this is $1 - r$: the models regenerate a correct continuation only about 20–45% of the time. With a resampled baseline the redundant hop drops to -0.3 to 4.3 points, as the proposition predicts, and AUC rises to 0.61–0.67. The remaining necessity of gold hops under resampling is small (-0.2 to 11 points) because the model re-derives a corrupted step from the context: resampling measures whether a step is *recoverable*, total-effect masking whether the written step is *used*, and the fixed-baseline estimator measures neither.

Table 4: Wrong-answer copy. A final hop stating a wrong relation is appended to a CLUTRR gold chain; we report how often the model's answer is that wrong relation, overall and on questions it answers correctly from the plain gold chain (PS= 1).

Model	PS of plain chain	follows wrong copy (all)	follows wrong copy | PS= 1
Q-0.5B	15.2	66.5 [58.9,74.1] (n=158)	70.8 [54.2,87.5] (n=24)
Q-1.5B	17.1	58.2 [50.6,66.5] (n=158)	63.0 [44.4,81.5] (n=27)
Smol-1.7B	26.6	56.3 [48.1,64.6] (n=158)	57.1 [42.9,71.4] (n=42)
Q-3B	27.8	46.2 [38.0,54.4] (n=158)	47.7 [34.1,61.4] (n=44)
Q-7B	33.1	22.9 [16.1,30.5] (n=118)	10.3 [2.6,20.5] (n=39)

R3: chains are read, not reasoned over. On CLUTRR the answer is a relation that must be composed from the hops and never appears in them. The total effect of a non-final gold hop is −1.5 to 1.8 points up to 1.7B, 7.3 [3.8, 11.0] at 3B and 1.3 [−3.4, 6.0] at 7B, an order of magnitude below the 52–77 points the same hops carry once one hop stating the answer is appended (T_{copy}, direct effect 0.9–1.7). Is this because the models cannot compose? The wrong-answer variant separates the two readings (table 4). Up to 3B the stated wrong answer is followed on 46–67% of questions, and *as often on questions the model solves from the plain chain* (48–71%) as on those it does not: copying overrides a solution the model can produce. Qwen2.5-7B follows it on 23% overall but on only 10% [3, 21] of the questions it solves, so at 7B a stated answer is checked against the model's own composition—yet even there the relational hops carry no measurable effect, i.e. the composition does not run through the chain. Chains that reach every gold-path entity also do not raise CoT accuracy (table 7). At every scale we test, the dependence on the chain tracks the readout the model can execute—copying a string—rather than the reasoning the chain encodes.

Table 5: Mediation on MuSiQue, `dist8`, n=500 (EM points; the NIE column is TE − NDE = TIE). Top: Qwen2.5-1.5B; bottom: Qwen2.5-3B. HGCD is omitted: these runs score only the full chain, under which it reduces to CoT.

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-5.6 [-9.2,-2.0]	-4.6 [-7.4,-1.6]	-1.0 [-3.4,1.6]
dist8	CoT	-2.6 [-6.2,1.2]	1.0 [-0.8,3.0]	-3.6 [-7.2,0.0]
dist8	CAD	-8.2 [-12.0,-4.0]	-7.0 [-10.2,-3.6]	-1.2 [-3.8,1.6]
dist8	AdaCAD	-7.6 [-11.2,-3.6]	-6.0 [-9.2,-2.8]	-1.6 [-4.2,1.2]
dist8	CAD-chain	-2.2 [-5.8,1.6]	1.6 [0.2,3.0]	-3.8 [-7.6,0.2]
dist8	Calibration (content-free)	-3.4 [-7.2,0.4]	0.0 [-1.2,1.2]	-3.4 [-7.2,0.6]
dist8	Direct × CoT (PoE)	-4.6 [-8.4,-0.8]	-0.2 [-2.4,2.0]	-4.4 [-8.0,-0.8]

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-5.8 [-10.0,-1.8]	-3.8 [-7.4,-0.2]	-2.0 [-4.4,0.4]
dist8	CoT	-5.6 [-9.6,-1.6]	-0.6 [-2.8,1.8]	-5.0 [-8.8,-1.0]
dist8	CAD	-7.4 [-11.4,-3.6]	-5.0 [-8.6,-1.6]	-2.4 [-5.0,0.2]
dist8	AdaCAD	-6.4 [-10.6,-2.6]	-4.0 [-7.8,-0.4]	-2.4 [-5.0,0.2]
dist8	CAD-chain	-5.6 [-10.0,-1.2]	-1.0 [-3.0,1.0]	-4.6 [-8.8,-0.6]
dist8	Calibration (content-free)	-6.4 [-10.4,-2.2]	-0.8 [-2.2,0.6]	-5.6 [-9.6,-1.6]
dist8	Direct × CoT (PoE)	-6.0 [-9.8,-2.2]	-0.4 [-2.8,2.0]	-5.6 [-9.2,-2.0]

R4: distractors act on chain-of-thought through the chain. On 500 MuSiQue questions, four extra distractor paragraphs change CoT exact match by −2.6 [−6.2, 1.2] (Qwen-1.5B) and −5.6 [−9.6, −1.6] (Qwen-3B). Feeding the original chain back under the distracted input removes the effect (NDE +1.0 and −0.6, both CIs covering 0); what remains goes through the regenerated chain (TIE −3.6 [−7.2, 0.0] and −5.0 [−8.8, −1.0]; table 5). Direct answering shows the opposite split (NDE −4.6 and −3.8, TIE $\geq$ −2.0), as does CAD (NDE −7.0 and −5.0). The pilot on 120 questions overestimated the magnitude (TE −9.2, TIE −10.0); the qualitative split replicates, the size is about 3–6 points. CoT is not more vulnerable than direct answering overall: the paired difference of total effects is +3.0 [−0.8, 6.8] and +0.2 [−3.8, 4.2]; nor is CAD (−2.6 [−6.0, 0.8], −1.6 [−4.4, 1.2]).

R5: a chain-level decoding probe cannot help, as R3–R4 predict. We formulate HGCD, which adds α times the average total effect of the chain's answer-relevant hops to each candidate's probability (appendix A). It can only amplify a dependence that exists and acts after the chain is generated. Consistently, it equals CoT on both datasets and models ($|\Delta| \leq 1.7$ points, $p \geq 0.6$). The rules that do beat CoT use no information from the chain: content-free label calibration [34] (+5.9 on CLUTRR for Qwen-1.5B; on MuSiQue with n=500, +4.0, p=0.002, and +5.4, $p < 0.001$, for 1.5B and 3B), which is itself fragile under a neutral-sentence placebo (-5.0 [-10.1, -0.4] for SmolLM2); and, at 1.5B, CAD on the direct readout (+8.6 over CoT, $p < 0.001$; not at 3B). The mixture-reader assumption behind our decoding propositions (appendix A) is not supported by these data.

Table 6: Main quantities for the two models with full decoding runs.

Quantity	qwen-1.5b	smollm2-1.7b
Content effect of chain, CLUTRR	0.8	2.0
Content effect, CLUTRR + answer hop (T_{copy}=1)	75.1	62.8
Content effect, MuSiQue	56.4	71.9
CLUTRR+copy non-final hops: Δ^{tot} / Δ^{nai}	76.3 / 1.3	68.5 / 1.1
MuSiQue non-final hops: Δ^{tot} / Δ^{nai}	66.3 / 0.7	72.8 / -3.6
AUC MuSiQue: total / direct	0.975 / 0.828	0.988 / 0.682
AUC CLUTRR+copy: total / direct	1.000 / 0.658	0.991 / 0.643
MuSiQue `dist8`, CoT: TE	-9.2	-7.5
NDE (chain fixed) / NIE (via chain)	0.8 / -10.0	1.7 / -9.2
MuSiQue `dist8`, direct answer: NDE	-4.2	-6.7
CLUTRR accuracy: calibration − CoT	5.9	3.4

Table 7: CLUTRR CoT accuracy by whether the model's own chain reaches every gold-path entity.

Method	gold-path chain (n=41)	other chain (n=197)
CoT	17.1 [7.3,29.3]	18.3 [13.2,23.9]
Self-consistency	9.8 [2.4,19.5]	23.4 [17.3,29.4]
CAD-chain	19.5 [7.3,31.7]	17.8 [12.7,23.4]
HGCD (naive mask)	22.0 [9.8,34.1]	23.4 [17.3,28.9]
HGCD	17.1 [7.3,29.3]	18.3 [13.2,23.9]

7 From diagnosis to decoding: execute the composition outside the model

R3–R5 locate the failure: the answer is read from the chain rather than computed over it, and distractors act through chain generation. Two decoding interventions follow, and we tested both against greedy free-form CoT and compute-matched self-consistency (SC), with every endpoint fixed in `REVIEW.md` before the run that tests it.

Cutting the distractor → chain path does not help. On MuSiQue we removed or reordered paragraphs before the chain was generated, using the model's calibrated attention from the question to each paragraph (calibrated as in ICR 3) or the model's own selection (table 10). Hard filtering keeps all gold paragraphs too rarely (top-4 recall 0.77–0.86 at 3B), and a lost gold paragraph halves

accuracy; reranking that keeps every paragraph is neutral; self-selection is worse. Nothing beats SC. The oracle that keeps only gold paragraphs gains 9–14 EM, so the distractor cost is real but not recoverable by these selectors at $\leq$ 3B. (Attention scores overflowed in fp16 for Qwen-1.5B and the sharded 7B; the 1.5B rows are from a corrected fp32 rerun, 7B attention results are excluded.)

Making the answer a function of the chain works. On CLUTRR the model only *extracts* facts ("X's *rel* is Y") and states each person's gender from the story's pronouns; the facts form a kinship graph, and a composition table induced from the *training* split (126 rules; 95% correct on gold chains) composes the relations along the path from the query's source to its target, falling back to CoT when no path exists (in the spirit of Faithful CoT 10, but with the extractor and the rules separated). The structural equations of the reasoning graph are thus executed rather than imitated. Chain-order parsing of the model's own CoT applied to only 4–6% of chains, and extraction without genders to 12–18%; diagnosis showed that most failures were missing inverse edges, which need the gender of the person they describe. With model-stated genders the method beats SC by 21–37 points on short chains at every scale, on the held-out validation split and on a CLUTRR task never used before (table 8); on unseen long chains (k=5–8) it helps from 3B (+9.5) and strongly at 7B (+24.0 [+17.0, +31.5], the pre-registered primary endpoint), but not at 1.5B, whose extraction is too incomplete.

The gain is caused by the composition, not by the re-presented facts. Holding the extraction fixed, we let the model answer from its own extracted facts instead of the rule table (table 9). On short chains this is no better than plain CoT (-3.2 to $+1.3$), while the rule table on the same facts is 13–35 points better at every scale, and 13 points better at 7B on long chains: the information needed to answer is present in the model's output, and the step that fails is composing it, exactly as R3 diagnosed.

Table 8: CLUTRR accuracy. Extract+rules: model-extracted facts and genders, composed by a rule table induced from the training split; Δ is Extract+rules SC_5 minus the better of CoT and SC_{10} (compute-matched), with paired bootstrap CI.

Evaluation set	Model	n	CoT	SC_{10}	Extract+rules	Extract+rules, SC_5	Δ vs. best baseline
Validation, k=2–4	1.5B	300	25.7	28.7	51.0	60.7	+32.0 [+26.0,+38.0]
	3B	300	36.3	38.0	64.3	70.0	+32.0 [+25.3,+38.3]
	7B	300	39.7	43.3	77.3	80.0	+36.7 [+31.0,+42.3]
Unseen task, k=2–4	1.5B	158	31.6	31.0	46.2	52.5	+20.9 [+12.0,+30.4]
	3B	158	30.4	32.3	54.4	59.5	+27.2 [+19.0,+35.4]
	7B	158	39.2	40.5	71.5	74.1	+33.5 [+25.3,+42.4]
Unseen task, k=5–8	1.5B	200	34.5	29.5	33.0	33.5	-1.0 [-8.5,+6.5]
	3B	200	26.0	29.0	32.0	38.5	+9.5 [+3.5,+15.5]
	7B	200	26.0	28.5	46.0	52.5	+24.0 [+17.0,+31.5]

8 Limitations and scope

Scale. Our models are 0.5–7B. R1 and R2 are identities whose magnitude we confirm at every scale. R3 changes at 7B in one respect (stated wrong answers are mostly vetoed) and not in the other (relational hops still carry no effect); whether composition over the chain appears at 14B and in other open families (OLMo-2) is the next experiment. The 7B run uses fewer chains and

Table 9: Ablation on the unseen CLUTRR task: same greedy extraction, composed by the LLM or by the rule table.

Chains	Model	n	CoT	LLM on own facts	Rules on own facts	Rules − LLM
k=2–4	1.5B	158	31.6	32.9	46.2	+13.3 [+5.1,+22.2]
	3B	158	30.4	28.5	54.4	+25.9 [+18.4,+34.2]
	7B	158	39.2	36.1	71.5	+35.4 [+27.2,+44.3]
k=5–8	1.5B	200	34.5	29.0	33.0	+4.0 [-3.5,+11.5]
	3B	200	26.0	27.0	32.0	+5.0 [-2.0,+12.5]
	7B	200	26.0	33.0	46.0	+13.0 [+5.0,+21.0]

Table 10: MuSiQue EM (n=500) when distractors are filtered before chain generation.

Distractors	Model	CoT	SC_5	Attn top-4	Attn rerank	Self-select	Gold only
4	1.5B	24.8	28.4	23.6	23.0	13.2	34.8
	3B	38.8	41.4	33.8	37.4	29.6	48.6
8	1.5B	22.2	25.0	22.0	22.2	13.2	31.4
	3B	31.6	35.6	27.4	33.6	24.8	45.6

no resampling estimators. **Sample size.** The estimator benchmark scores about 2,800 hops per model and mediation uses 500 questions; the CLUTRR decoding comparisons use 238 questions, so accuracy differences below about 6 points there are not resolved. The wrong-copy follow rate on solvable questions rests on 24–44 questions per model (CIs in table 4). **Assumptions.** assumption 1 idealises the reader; the data match its prediction. The mixture assumption used for the decoding probe does not hold (R5). **Masking.** Subtree masking inserts several mask tokens. Counterfactual replacement, the raw log-probability effect and the volume-matched marker control agree with it in magnitude, but the marker control is not inert at 0.5B (R1). **Parsing.** The DRG is built from explicit mentions. On the models' own CLUTRR chains, format success is 100%, gold-path entities are reached in 57% (Qwen) / 60% (SmolLM2) of hops, and the DRG is fully connected in 25% / 36% of chains; this affects only the decoding probe, since the estimator benchmark uses gold chains. **Candidate sets.** Effects on MuSiQue are measured on the gold answer's probability over a candidate set (gold recall 56% before adding the model's strings); the raw log-probability version, which needs no candidate set, gives the same picture (table 3).

9 Conclusion

With interventions whose ground truth is known, two common ways of measuring the importance of a reasoning step turn out to be biased in predictable ways: single-step masking measures a direct effect that vanishes on bridge chains, and fixed-baseline resampling adds the generator's failure rate. Correctly measured, models from 0.5B to 7B depend on their chains when the answer can be read from them, not when it must be computed; small models copy a stated answer even against their own solution, and 7B vetoes it without composing through the chain either. Distractors hurt chain-of-thought through the chain it regenerates. Decoding rules that act on the chain's readout cannot repair either; executing the composition the chain describes, outside the model, can.

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Algorithm 1 HGCD (L2) for one question

$y \leftarrow$ greedy chain from the model (optionally with L1 contrast on result tokens)
Parse hops; build DRG; compute relevant set $\mathcal{R}(y)$
$p \leftarrow \pi(\cdot \mid Q, P, y)$ over $\mathcal{A}$ ▷ one forward + KV-cached candidates
for $j \in \mathcal{R}(y)$ **do**
    $p_j \leftarrow \pi(\cdot \mid Q, P, y_j^{\text{tot}})$
end for
return $\arg\max_{a \in \mathcal{V}_\beta} (1+\alpha) p(a) - \alpha \,\text{mean}_j\, p_j(a)$

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A The decoding probe in full

Answer-level rule (L2). With uniform weights $w_j = 1/|\mathcal{R}(y)|$ on relevant hops and a plausibility set $\mathcal{V}_\beta = \{a : \pi(a \mid y) \geq \beta \max_{a'} \pi(a' \mid y)\}$ [9],

$$\begin{aligned} \hat{a} &= \arg\max_{a \in \mathcal{V}_\beta} s_\alpha(a), \\ s_\alpha(a) &= (1+\alpha)\,\pi(a \mid y) - \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \pi(a \mid y_j^{\text{tot}}) = \pi(a \mid y) + \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \Delta_j^{\text{tot}}(a). \end{aligned} \tag{1}$$

The score is the answer probability plus α times the average total effect of the answer-relevant hops on that answer. Redundant hops ($j \notin \mathcal{R}(y)$) are never contrasted.

Evidence-level rule (L1). While emitting the *result* tokens of a hop, we contrast the real context P with a type-matched counterfactual $\tilde{P}$ of identical length and entity types in which the relational content differs (on CLUTRR: names permuted within each gender group, so every sentence and entity type is kept but who-is-related-to-whom changes; we evaluate L1 on CLUTRR only, where $\tilde{P}$ is exactly type-matched): $t^\star = \arg\max_{t \in \mathcal{V}_\beta} (1+\alpha)\pi(t \mid P, \cdot) - \alpha\pi(t \mid \tilde{P}, \cdot)$. Unlike an empty context, $\tilde{P}$ is in-distribution and controls for length and type cues.

Remark 1 (What each level can fix)**.** *L2 makes the answer depend* more *on the chain; it therefore repairs* readout *errors (chain correct, answer wrong because a shortcut or distractor dominates the readout) but cannot repair a wrong chain—it may even commit harder to one. Chain errors are the target of L1, which acts while the hops are produced.*

Cost. L2 needs $1 + |\mathcal{R}(y)|$ prefix forwards and a KV-cached pass over the candidate tokens; no extra generation. L1 doubles the forward cost of result tokens only.

Choice of α. α is selected on the CLUTRR validation split and frozen (ties broken toward the smaller α, so HGCD reduces to CoT absent evidence); an adaptive variant sets $\alpha \cdot \mathrm{JSD}(p, \bar{p}_j)$ as in AdaCAD [29].

Propositions for the decoding probe

These propositions rest on a mixture-reader assumption that our data do not support (R5); we keep them because they make the probe's intended mechanism explicit.

Assumption 2 (Mixture reader)**.** $\pi(a \mid Q, P, y) = \lambda_c \pi_c(a \mid y, P_{\text{sup}}) + \lambda_s \pi_s(a \mid Q) + \lambda_d \pi_d(a \mid P_{\text{dist}})$*, with λ's non-negative, summing to one, and invariant to interventions on hops.*

Proposition 3 (Exact cancellation in probability space)**.** *Under assumption 2,*

$$s_\alpha(a) = \pi(a \mid y) + \alpha\lambda_c \sum_j w_j \big[\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})\big],$$

i.e. the shortcut π_s and distractor π_d components cancel in the contrast for every α. The log-space analogue does not cancel them.

Proposition 4 (CAD amplifies distractor paths)**.** *Under assumption 2 applied to direct answering, with no-context distribution $\pi_s(a \mid Q)$, CAD's score is $(1{+}\alpha)\log\big[\lambda_c\pi_c{+}\lambda_s\pi_s{+}\lambda_d\pi_d\big](a){-}\alpha\log\pi_s(a \mid Q)$. For any $\alpha > 0$ and any gold answer g, a distractor answer a_d with $\pi_d(a_d) = 1$ is preferred by CAD whenever $\pi_s(a_d \mid Q)$ is small enough, even if $\pi(g \mid Q, P) > \pi(a_d \mid Q, P)$. In contrast, HGCD's contrast term does not depend on π_d at all (proposition 3).*

Prediction. Under distractor injection, CAD's accuracy degrades more than HGCD's, i.e. $|\mathrm{ATE}_{\text{distract}}|$ is larger for CAD.

Proposition 5 (PNS limit)**.** *Interpret $\pi(a \mid y_j^{\text{tot}})$ as $P(A{=}a \mid \mathrm{do}(H_j{=}\perp))$ and assume exogeneity and monotonicity as in Yu et al. [32]. Then $\Delta_j^{\text{tot}}(a) = \mathrm{PNS}_j(a)$ and $\lim_{\alpha\to\infty} s_\alpha(a)/\alpha = \sum_{j\in\mathcal{R}(y)} w_j \,\mathrm{PNS}_j(a)$. Hence, within $\mathcal{V}_\beta$, HGCD interpolates between the most probable answer ($\alpha{=}0$) and the answer for which the relevant hops are, on average, most necessary and sufficient.*

B Proofs

Proposition 1. y_j^{nai} changes only h_j and keeps $h_{j+1:k}$ at their observed values, so Δ_j^{nai} is by definition the controlled direct effect $\pi(a \mid h_j, h_{-j}) - \pi(a \mid \perp, h_{-j})$ [15]. Under assumption 1, $\pi(a \mid h_{1:k}) = g(a \mid h_{j+1:k})$ for $j < k$, so both terms equal $g(a \mid h_{j+1:k})$ and the difference is zero. y_j^{tot} additionally changes $h_{j+1:k}$ (their results are masked), so g is evaluated at a different argument and the difference is unconstrained. On compositional chains π reads the relation of every hop, so removing h_j alone changes the composed relation. □

Proposition 3. Substituting assumption 2 into eq. (1), with $\sum_j w_j = 1$ and π_s, π_d unaffected by interventions on hops: $s_\alpha(a) = (1{+}\alpha)[\lambda_c\pi_c(a \mid y) + \lambda_s\pi_s + \lambda_d\pi_d] - \alpha\sum_j w_j[\lambda_c\pi_c(a \mid y_j^{\text{tot}}) + \lambda_s\pi_s + \lambda_d\pi_d] = \pi(a \mid y) + \alpha\lambda_c\sum_j w_j[\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$. In log space, $\log(\lambda_c u + r) - \log(\lambda_c v + r)$ depends on $r = \lambda_s\pi_s + \lambda_d\pi_d$ unless $u = v$, so the shortcut terms do not cancel. □

Proposition 4. Let $m(a) = \pi(a \mid Q, P)$. CAD prefers a_d over g iff $(1+\alpha)\log\frac{m(a_d)}{m(g)} > \alpha \log\frac{\pi_s(a_d)}{\pi_s(g)}$. The left side is bounded below by $(1+\alpha)\log\frac{\lambda_d}{m(g)}$ since $m(a_d) \geq \lambda_d$, while the right side tends to $-\infty$ as $\pi_s(a_d) \to 0$. For HGCD, both terms of eq. (1) condition on the same P, so $\lambda_d \pi_d$ enters both and cancels (proposition 3). □

Proposition 5. Under exogeneity and monotonicity, Lemma 2 of Yu et al. [32] gives PNS = $P(A{=}a \mid \mathrm{do}(S)) - P(A{=}a \mid \mathrm{do}(S'))$. With $S = y$ and $S' = y_j^{\mathrm{tot}}$ this is $\Delta_j^{\mathrm{tot}}(a)$. Dividing eq. (1) by α and letting $\alpha \to \infty$ removes the $\pi(a \mid y)/\alpha$ term. □

C Implementation and Reproducibility

Candidate log-probabilities are computed with one prefix forward and a KV-cached pass over the candidate tokens (verified against full recomputation to 2×10^{-5}). Prompts are two-shot, hop-structured; the second CLUTRR shot uses a relation outside the label set to avoid leaking label statistics. All interventions are seeded functions of the example id. The Kaggle notebook embeds the full package and writes every table in this paper.

D SmolLM2-1.7B tables

Estimator	clutrr		clutrr_copy		musique	
	all	PS=1	all	PS=1	all	PS=1
Mask hop only (direct)	0.540	0.516	0.643	0.645	0.682	0.696
Mask hop + DRG descendants (total)	0.559	0.559	0.991	0.996	0.988	0.992
Replace result, keep downstream (direct)	0.503	0.474	0.651	0.643	0.658	0.658
Replace result, propagate (total)	0.532	0.494	0.996	1.000	0.987	0.993
DRG path indicator (structure only)	1.000	1.000	0.500	0.500	0.994	0.993
Placebo rewrite (control)	0.548	0.548	0.388	0.386	0.513	0.513
Mask token, content kept (control)	0.508	0.480	0.535	0.544	0.500	0.500

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-5.8 [-11.7,0.0]	-6.7 [-11.7,-1.7]	0.8 [-1.7,4.2]
dist8	CoT	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	CAD	-4.2 [-9.2,0.8]	-5.0 [-9.2,-1.7]	0.8 [-1.7,4.2]
dist8	AdaCAD	-3.3 [-8.3,0.8]	-5.0 [-10.0,-0.8]	1.7 [-1.7,5.0]
dist8	CAD-chain	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	Calibration (content-free)	-10.8 [-17.5,-3.3]	-1.7 [-5.0,1.7]	-9.2 [-15.8,-2.5]
dist8	Direct × CoT (PoE)	-7.5 [-14.2,0.0]	-0.8 [-4.2,1.7]	-6.7 [-13.3,0.0]
dist8	HGCD (no graph)	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
dist8	HGCD (log-space)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	HGCD (adaptive α)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	**HGCD**	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
order	Direct (no CoT)	-2.5 [-7.5,2.5]	-0.8 [-4.2,2.5]	-1.7 [-5.8,2.5]
order	CoT	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	CAD	-0.8 [-5.0,3.3]	0.0 [-4.2,4.2]	-0.8 [-4.2,2.5]
order	AdaCAD	-0.8 [-4.2,2.5]	-0.8 [-4.2,2.5]	0.0 [-3.3,3.3]
order	CAD-chain	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	Calibration (content-free)	0.0 [-7.5,7.5]	-0.8 [-2.5,0.0]	0.8 [-6.7,8.3]
order	Direct × CoT (PoE)	1.7 [-5.8,9.2]	-0.8 [-4.2,1.7]	2.5 [-4.2,10.0]
order	HGCD (no graph)	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]
order	HGCD (log-space)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	HGCD (adaptive α)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	**HGCD**	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]